

Second law of motion

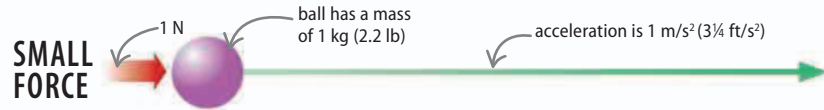
The second law of motion states that when a force acts on an object, it will tend to move in the direction of the force. The larger the force on an object, the greater its acceleration will be. The more massive an object is, the greater the force needed to accelerate it.

product of mass and acceleration (newtons (N)) amount of matter an object contains (kilograms) increase in velocity over time (meters per second per second (m/s^2))

$$\text{Force} = \text{Mass} \times \text{Acceleration}$$

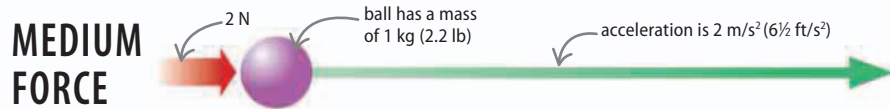
▷ Small mass, small force

A force of 1 N acting on a mass of 1 kg (2.2 lb) will produce an acceleration of 1 m/s^2 ($3\frac{1}{4} \text{ ft/s}^2$)—velocity increases by 1 m ($3\frac{1}{4} \text{ ft}$) per second every second.



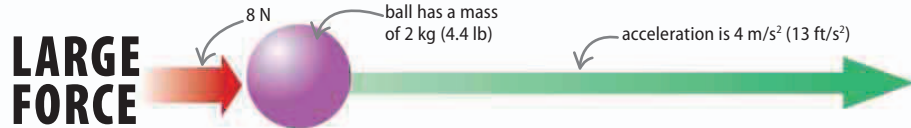
▷ Small mass, medium force

A force of 2 N acting on a mass of 1 kg (2.2 lb) will produce an acceleration of 2 m/s^2 ($6\frac{1}{2} \text{ ft/s}^2$).



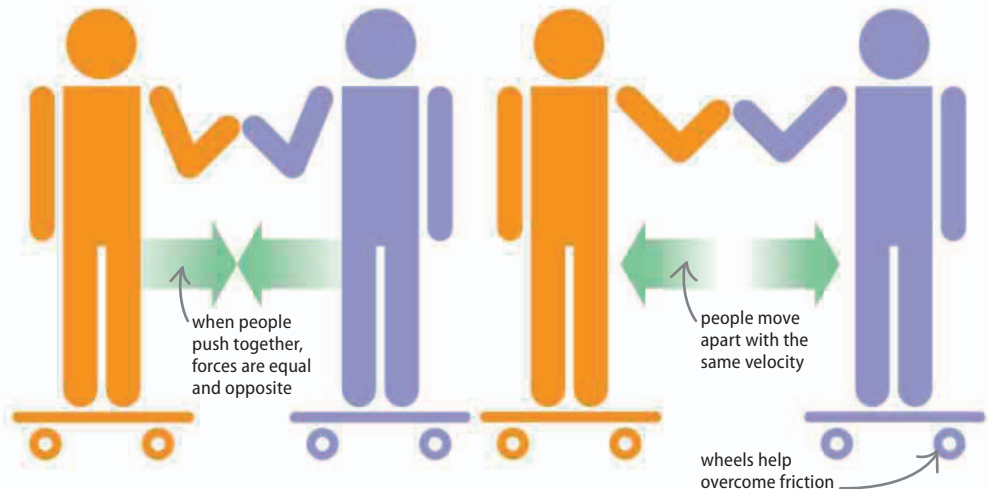
▷ Double mass, large force

A force of 8 N acting on a mass of 2 kg (4.4 lb) will produce an acceleration of 4 m/s^2 (13 ft/s^2).



Third law of motion

The third law of motion states that any object will react to a force applied to it. The force of reaction is equal and acts in an opposite direction to the original force that produces it. In the diagram, two people of the same mass are each standing on a skateboard, which reduces friction. When they push together (action), the result (reaction) will be that they move in the opposite direction from each other.



△ Action

The third law works when a force acts between two objects. Even if the second person does not make any effort to push back, her or his body will always react to the force in the same way.

△ Reaction

The forces between the two people are equal and opposite as they push away from each other on their skateboards. The masses are equal, so they move away from each other at the same velocity.